Culture-independent detection of *Staphylococcus aureus* using multiple displacement amplification and CRISPR-Cas9 coupled with Nanopore sequencing

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Introduction

Staphylococcus aureus is a high-priority healthcare-associated pathogen, with methicillin-resistant strains (MRSA) causing more than 70,000 severe infections and 9,000 deaths annually in the U.S.[1]

Culture-independent detection in environmental microbiomes remains challenging because low-biomass samples are especially vulnerable to contamination, and low target abundance can limit sequencing recovery[2,3].

In addition, methods that increase DNA yield or enrich target-specific signal may improve sequencing performance in complex samples[3,4].

Objective

To assess whether using MDA and CRISPR-Cas9 enrichment with Nanopore sequencing improves the culture-independent detection of *S. aureus* in complex environmental microbiome samples at 10^{-2} and 10^{-3} spike-in levels.

Materials and Methods

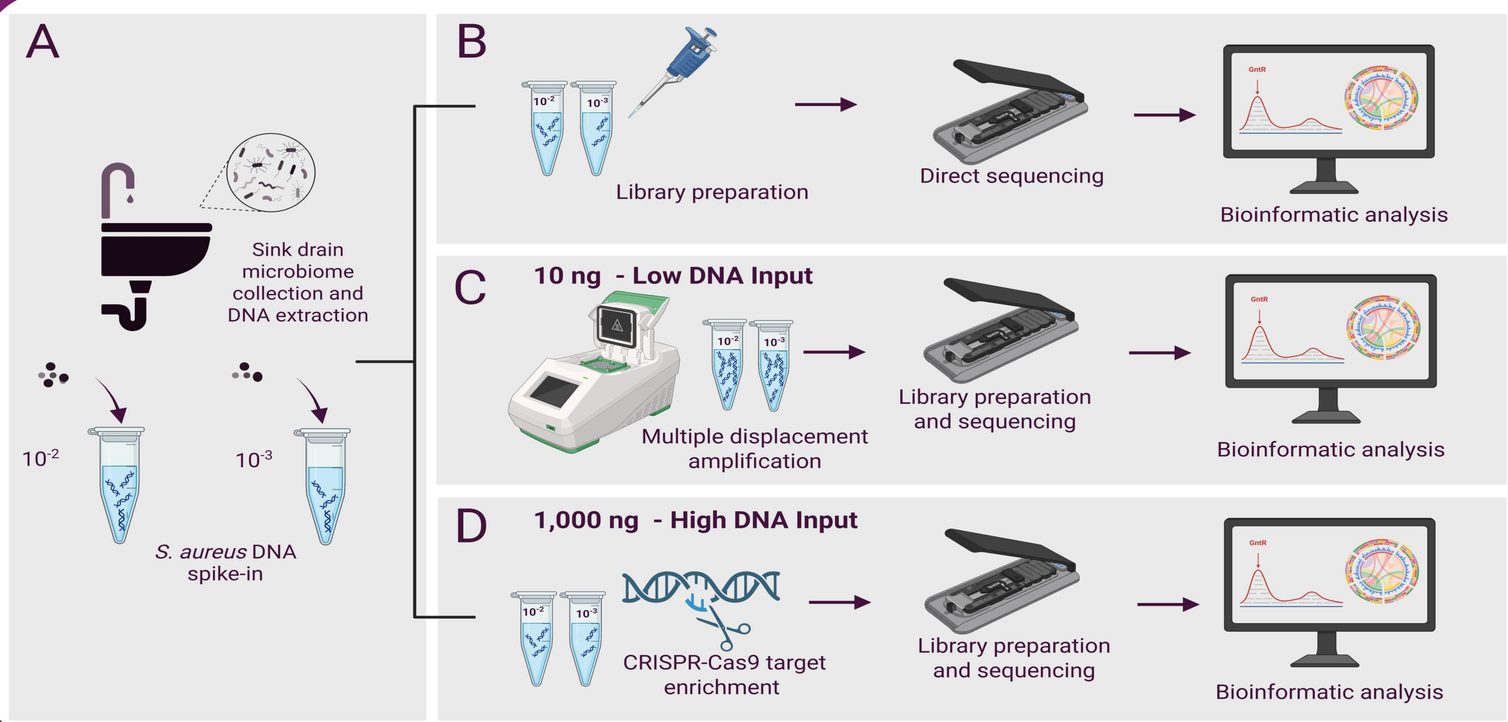

Figure 1. Experimental workflow used to evaluate culture-independent detection of *S. aureus* in environmental microbiome samples. **(A)** Environmental microbiome was collected from sink drain swabs, extracted using the DNeasy PowerBiofilm kit, and spiked with *S. aureus* ATCC 12600 genomic DNA at 10^{-2} and 10^{-3} ratios. **(B)** For direct sequencing, 1,000 ng total DNA was processed by library preparation and Nanopore sequencing. **(C)** For the low-input workflow, 10 ng total DNA was first subjected to multiple displacement amplification (MDA), and the amplified product was adjusted to 1,000 ng for library preparation and sequencing. **(D)** For the high-input workflow, 1,000 ng total DNA was processed by CRISPR-Cas9 target enrichment of the *S. aureus gntR* locus before library preparation and sequencing. Libraries were prepared using the ONT Native Barcoding Kit V14 and sequenced on R10.4.1 flow cells using a MinION Mk1D for 16 h. Reads were basecalled and demultiplexed with Dorado/MinKNOW, mapped to the *S. aureus* ATCC 12600 reference genome using Minimap2, and analyzed using normalized *S. aureus* signal, breadth of genome coverage, and *gntR* on-target enrichment metrics.

Results and Discussion

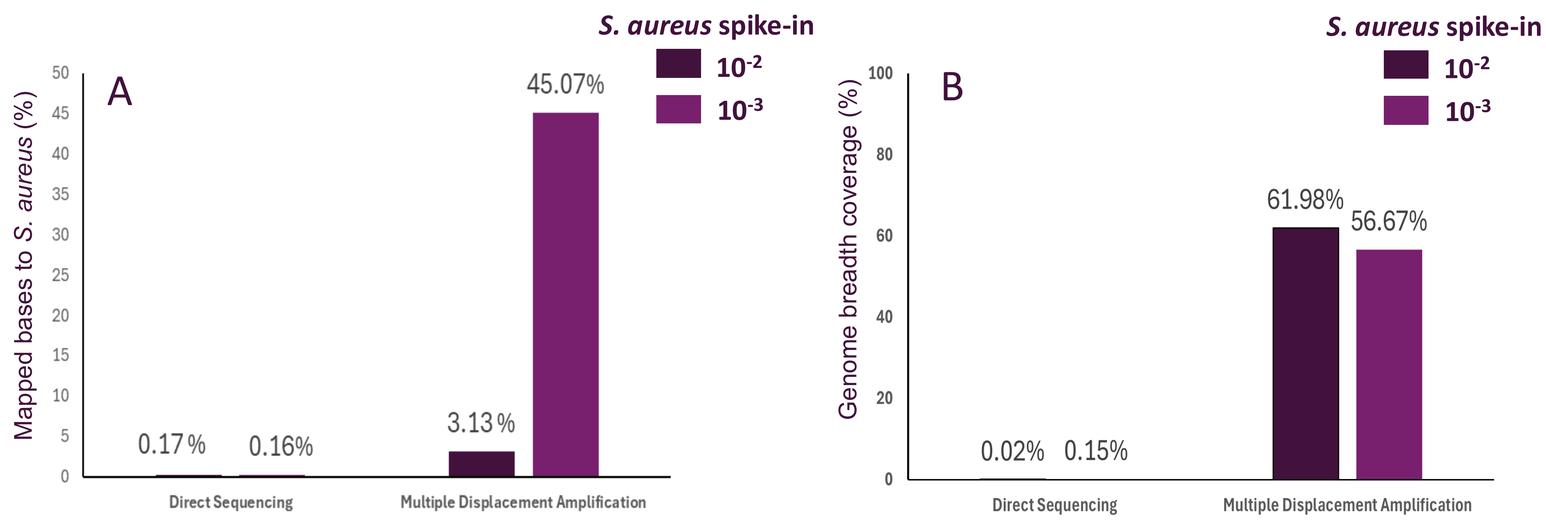

Figure 2. Effect of multiple displacement amplification (MDA) on *S. aureus* recovery in environmental microbiome samples using 10 ng total DNA input. **(A)** MDA increased the proportion of sequenced bases mapped to the *S. aureus* genome, from 0.17% and 0.16% in direct sequencing samples to 3.13% and 45.07% in MDA samples at 10^{-2} and 10^{-3} spike-in levels, respectively. **(B)** MDA also increased the breadth of genome coverage, from 0.02% and 0.15% to 61.98% and 56.67% at 10^{-2} and 10^{-3}, respectively.

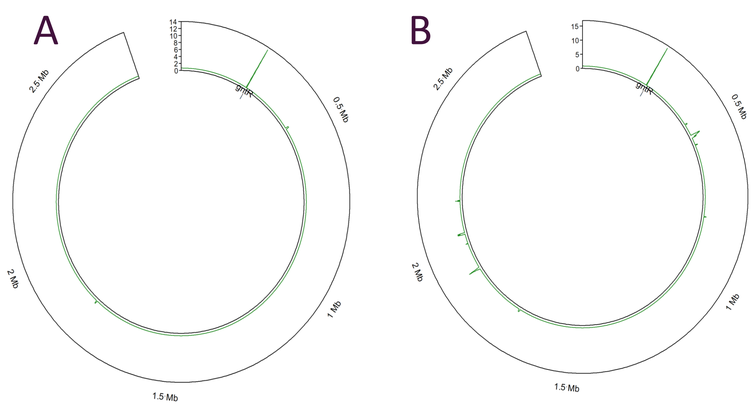

Figure 3. Circular genome plots of CRISPR-Cas9 enriched high-input samples showing localized peak at the *S. aureus gntR* target. **(A)** 10^{-2} spike-in sample, with 14 reads mapped to *gntR*. **(B)** 10^{-3} spike-in sample, with 17 reads mapped to *gntR*.

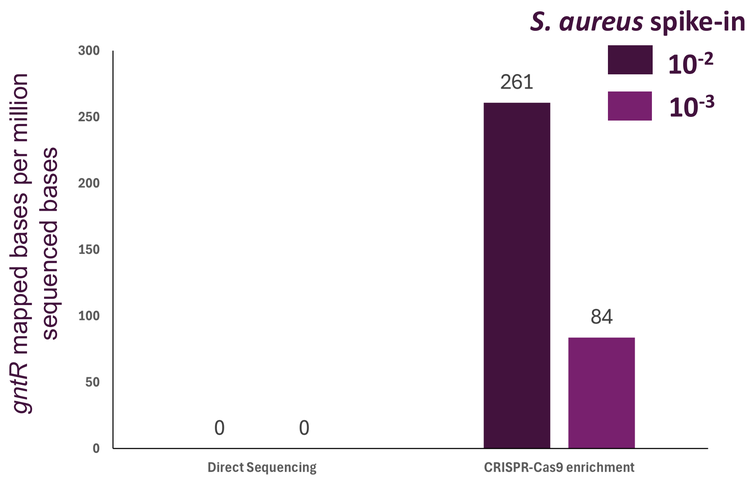

Figure 4. CRISPR-Cas9 increased recovery of the *S. aureus gntR* target in high-input environmental microbiome samples. No *gntR* signal was detected by direct sequencing, whereas CRISPR-Cas9 enrichment yielded 261 and 84 mapped bases per million sequenced bases at 10^{-2} and 10^{-3} spike-in levels, respectively.

Significance

This proof-of-concept study supports the development of a sequencing-based method for the detection of *S. aureus* in environmental samples, enabling applications in surveillance, infection prevention, and genomic characterization such as AMR and virulence gene profiling.

Conclusions

In this proof-of-concept experiment, MDA was an effective strategy for improving overall detection and genome-wide recovery of *S. aureus* in low-input environmental microbiome samples. Furthermore, CRISPR-Cas9 primarily increased signal at the *gntR* locus, making it useful for target confirmation.

Future Directions

Increase experimental replication to enable more robust comparison of workflow performance. Validate the workflow using environmental samples naturally contaminated with *S. aureus*.

References

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(4) Cottingham H, Judd LM, Wisniewski JA, Wick RR, Stanton TD, Vezina B, *et al*. Targeted sequencing of *Enterobacterales* bacteria using CRISPR-Cas9 enrichment and Oxford Nanopore Technologies. mSystems. 2025.

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